

Input Tokens

t	0	1	2	3	4	5
	3	3	0	3	0	2
	2	0	3	2	3	1
	1	0	1	1	1	3
	3	1	2	3	1	1
	2	3	2	2	1	2
	↓	↓	↓	↓	↓	↓

Multi-head Self-Attention

Head 0

Linear

					Wq	
1	-1	1	0	0		→
0	0	-1	0	0		→
1	-1	1	0	1		→

					Wk	
1	0	0	0	0		→
0	1	0	1	1		→
0	0	0	1	0		→

					Wv	
0	1	0	0	0		→
1	0	0	0	0		→
0	0	0	1	-1		→

					Q
2	3	-2	2	-2	4
-1	0	-1	-1	-1	-3
4	6	0	4	-1	6

					K
3	3	0	3	0	2
7	4	7	7	5	4
3	1	2	3	1	1

					V
2	0	3	2	3	1
3	3	0	3	0	2
1	-2	0	1	0	-1

Scaled Dot-Product

t

012345

Q

2	3	-2	2	-2	4
-1	0	-1	-1	-1	-3
4	6	0	4	-1	6

K^T

0	3	7	3
1	3	4	1
2	0	7	2
3	3	7	3
4	0	5	1
5	2	4	1

K^T * Q / sqrt(dk)

↓↓↓↓↓↓

→	6.35	15.6	-7.5	6.35	-9.2	5.2
→	3.46	8.66	-5.8	3.46	-6.4	3.46
→	0.58	6.93	-4	0.58	-5.2	-5.2
→	6.35	15.6	-7.5	6.35	-9.2	5.2
→	-0.6	3.46	-2.9	-0.6	-3.5	-5.2
→	2.31	6.93	-4.6	2.31	-5.2	1.15

V

2	0	3	2	3	1
3	3	0	3	0	2
1	-2	0	1	0	-1

Softmax

	0	1	2	3	4	5
0	0.48	0.5	0.01	0.48	0	0.46
1	0.03	0	0.04	0.03	0.04	0.08
2	0	0	0.2	0	0.12	0
3	0.48	0.5	0.01	0.48	0	0.46
4	0	0	0.64	0	0.71	0
5	0.01	0	0.11	0.01	0.12	0.01

V'

↓↓↓↓↓↓

→	1.94	2	2.65	1.94	2.63	1.83
→	2.99	3	0.37	2.99	0.38	2.99
→	0.9	1	-0.2	0.9	-0.2	0.74

Head 1

Linear

1	-1	0	1	1	→
0	1	-1	1	-1	→
1	1	0	0	0	→

1	0	1	-1	1	→
1	0	0	1	0	→
1	1	0	1	1	→

1	0	-1	0	-1	→
0	1	0	0	0	→
0	1	1	0	0	→

Wq

Wk

Wv

Q

6	7	1	6	-1	4
2	-2	2	2	2	-3
5	3	3	5	3	3

K

3	5	1	3	1	6
6	4	2	6	1	3
10	7	7	10	5	6

V

0	0	-3	0	-2	-3
2	0	3	2	3	1
3	0	4	3	4	4

Scaled Dot-Product											
						t					
						0	1	2	3	4	5
						Q					
6	7	1	6	-1	4						
2	-2	2	2	2	-3						
5	3	3	5	3	3						
						K^T * Q / sqrt(dk)					
						↓	↓	↓	↓	↓	↓
0	3	6	10	→	46.2	22.5	26	46.2	22.5	13.9	
1	5	4	7	→	42.1	27.7	19.6	42.1	13.9	16.7	
2	1	2	7	→	26	13.9	15	26	13.9	11	
3	3	6	10	→	46.2	22.5	26	46.2	22.5	13.9	
4	1	1	5	→	19.1	11.5	10.4	19.1	9.24	9.24	
5	6	3	6	→	41.6	31.2	17.3	41.6	10.4	19.1	
						Softmax					
						0	1	2	3	4	5
0	0.49	0	0.5	0.49	0.5	0					
1	0.01	0.03	0	0.01	0	0.09					
2	0	0	0	0	0	0					
3	0.49	0	0.5	0.49	0.5	0					
4	0	0	0	0	0	0					
5	0	0.97	0	0	0	0.9					

V	0	0	-3	0	-2	-3
V'	↓	↓	↓	↓	↓	↓
→	-0	-2.9	-0	-0	-0	-2.7
→	1.98	0.97	2	1.98	2	0.92
→	2.98	3.88	3	2.98	3	3.63

Head 2

Linear

Wq	0	-1	0	0	0
Wk	0	1	-1	0	0
Wv	-1	1	1	0	1
→	0	1	1	1	0
→	-1	0	-1	0	1
→	0	1	-1	1	-1
→	-1	0	1	1	1
→	-1	1	1	0	1
→	0	1	0	1	1
→	0	0	1	0	0

Q	-2	0	-3	-2	-3	-1
K	1	0	2	1	2	-2
V	2	0	6	2	5	4
→	6	1	6	6	5	5
→	-2	0	1	-2	0	-3
→	1	1	-4	1	-4	-3
→	3	1	5	3	3	4
→	7	4	7	7	5	4
→	1	0	1	1	1	3

Scaled Dot-Product

Scaled Dot-Product

t

0

1

2

3

4

5

Q

-2	0	-3	-2	-3	-1
6	1	6	6	5	5
-2	0	1	-2	0	-3

K^T

1	1	3
0	1	1
2	-4	5
1	1	3
2	-4	3
-2	-3	4

K^T * Q / sqrt(dk)

↓

↓

↓

↓

↓

↓

-1.2	0.58	3.46	-1.2	1.15	-2.9
2.31	0.58	4.04	2.31	2.89	1.15
-22	-2.3	-14	-22	-15	-21
-1.2	0.58	3.46	-1.2	1.15	-2.9
-20	-2.3	-16	-20	-15	-18
-13	-1.7	-4.6	-13	-5.2	-14

Softmax

	0	1	2	3	4	5
0	0.03	0.31	0.26	0.03	0.13	0.02
1	0.94	0.31	0.47	0.94	0.74	0.97
2	0	0.02	0	0	0	0
3	0.03	0.31	0.26	0.03	0.13	0.02
4	0	0.02	0	0	0	0
5	0	0.03	0	0	0	0

V'

↓

↓

↓

↓

↓

↓

0.12	1.56	1.06	0.12	0.52	0.07
4.18	5.94	5.59	4.18	4.78	4.1
0.06	0.75	0.53	0.06	0.26	0.03

V

2	0	6	2	5	4
7	4	7	7	5	4
1	0	1	1	1	3

Concat

Head	0	1	2
V'	1.94	2	2.65
→	2.99	3	0.37
→	0.9	1	-0.2
→	-0	-2.9	-0
→	1.98	0.97	2
→	2.98	3.88	3
→	0.12	1.56	1.06
→	4.18	5.94	5.59
→	0.06	0.75	0.53

Linear

Linear									Wo			↓	↓	↓	↓	↓	↓
0	0	1	0	0	1	-1	-1	-1	→	Output	-0.5	-3.4	-4.3	-0.5	-2.8	0.17	
1	1	1	0	1	1	1	1	1	→	15.1	19.1	15	15.1	13.4	14.3		
1	0	-1	1	0	0	1	0	0	→	1.14	-0.3	3.88	1.14	3.35	-1.5		
1	0	0	0	0	-1	1	0	0	→	-0.9	-0.3	0.72	-0.9	0.15	-1.7		
-1	1	1	1	0	0	-1	1	0	→	5.99	3.47	2.07	5.99	1.81	3.24		

Output Tokens



